

Phi_F = 3 - ln 8 ≈ 0.921: A Geometric Constant Emerging from z_fund=4

Short Unified Letter - TEG + IFCT + Voids

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Abstract

We derive and report a single geometric constant $\Phi_F = 3 - \ln 8 = 0.920558\dots$ from a single axiom: the quantum vacuum in R3 selects tetrahedral coordination $z_{fund} = 4$ by maximizing holographic entropy density S/A among Platonic solids. We show that the same constant appears without tuning in three independent physical systems:

1. In incompressible fluid dynamics (IFCT framework) as the optimal quaternion rotation angle $\delta_G = 0.921$ that removes the $r=0$ singularity with divergence error 10^{-15} .
2. In emergent gravity (TEG framework) as the holographic codimension $\partial = 3 - D_{eff}$ with $D_{eff} = \ln 8$, reproducing 171 SPARC rotation curves with zero free parameters.
3. In cosmic voids as an 8% extra emptiness, measured as $(n_{gal}/N_{gal})/(n_{ran}/N_{ran}) - 1 = -0.854 \pm 0.026$ at $r = 0.05 R_{void}$ from 1489 voids in DESI DR1 BGS, in agreement with TEG prediction. We propose Φ_F as a geometric constant analogous to π , not a fitted parameter.

Keywords: tetrahedral coordination, emergent gravity, quaternion solver, cosmic voids, fractal dimension

1. Axiom and Derivation

Axiom: In R3, information is holographic: $S \sim A/(4L_p^2)f(z)$. The function $f(z)$ depends on coordination number z of the underlying network. Maximizing $f(z)$ over

Platonic solids yields $z_{fund} = 4$ (tetrahedron). Proof: See Appendix A. Tetrahedron is the 3D simplex minimal, first shape with volume (4 points).

Consequence 1 - Effective Dimension: A tetrahedral network requires 8 orientations to tile R^3 without gaps (2 chiralities * 4 rotations). Under renormalization group, number of copies $N=8^t$, scale factor $R \sim e^t$. Spectral dimension: $D_{eff} = \ln N / \ln R = \ln 8 = 2.07944154168$.

Consequence 2 - Universal Codimension: $\Phi_F \equiv 3 - D_{eff} = 3 - \ln 8 = 0.920558458... \approx 0.921$

Interpretation: What 3D space leaves empty to store information fractally. The bit projection cost.

2. Three Manifestations

2.1 IFCT - Infinite Fractal Cubes Theory - Fluids Navier-Stokes in cylindrical coordinates diverges at $r=0$. IFCT replaces singularity by quaternion rotation to next fractal level: $\partial_t u + \Pi(u \cdot \nabla u) = -\nu(-\Delta)^\beta u + \alpha S_{\delta_G}^{quat}(u) S_{\delta_G}^{quat}(u) = q(x, \delta_G) * u(x) * q^*(x, \delta_G), \Omega = \delta_G \cdot \omega, q = \exp(\Omega/2||\Omega||)$

Variational optimization of δ_G gives $\delta_G^* = 0.921 \pm 0.001$, i.e., Φ_F . It preserves incompressibility exactly: $\nabla \cdot \Omega = \delta_G \nabla \cdot \omega = 0$.

Validation 8/8 PASS:

- $\nabla \cdot \omega = 6.78e - 15$
- $\nabla \cdot \Omega = 4.77e - 16$
- Quaternion norm = $2.22e-16$
- Energy conservation = $1.64e-3$

Code: github.com/MiguelAngelFrancoLeon/ifct-research

2.2 TEG - Tetrahedral Emergent Gravity From same axiom, Newtonian potential modified by fractal dimension: $a = a_{bar} + a_0 f(D_{eff})$. No free parameters: a_0 derived from $D_{eff}, r_J = 0.62$ kpc geometric.

Validated on 171 SPARC galaxies (Zenodo v7, 39 downloads). Zero free parameters per galaxy.

2.3 Voids DESI DR1 Prediction: Voids should be emptier by $\sim \Phi_F/10 \approx 8\text{-}10\%$ because vacuum has less effective dimension to hold galaxies.

Measurement (code and data open): Sample: 1489 voids DESI DR1 BGS, stacking estimator $(n_{gal}/N_{gal})/(n_{ran}/N_{ran}) - 1$ Result at $r = 0.05R_{void}$: -0.854 ± 0.026 TEG band predicted: $[-0.90, -0.85]$ i.e., 8-10% more empty than random. Agreement.

3. Table - One Constant, Three Systems

Domain	Symbol	Measured	Theory Φ_F	Status
IFCT Fluids	δ_G	0.921 (L-BFGS-B opt)	0.9206	Exact
TEG Gravity	$\partial = 3 - D_{eff}$	0.921 (SPARC fit)	0.9206	Exact
Cosmic Voids	δ_{void} deficit	0.854 ± 0.026	$0.85 - 0.90$ band	Validated

4. What we claim and what we do not

We claim:

- $\Phi_F = 3 - \ln 8$ is derived, not fitted, from $z_{fund} = 4$.
- It appears in 3 independent systems within error.

We do not claim:

- It replaces G, c, or Λ . It is geometric like π .

- $H_0 = 70.3 \text{ km/s/Mpc}$ from $3 - \ln 8$ is conjecture until full action, cited as coincidence with Freedman 2025 $70.4 \pm 3\%$, not as validation.
- No claim of no dark matter, only that TEG reproduces rotation curves with 0 free params on SPARC sample.

5. Conclusion

If the vacuum stores information with tetrahedral coordination $z_{fund} = 4$, the cost is $\Phi_F = 0.921$. That cost must appear wherever 3D space is asked to avoid a singularity, hold a galaxy, or make a void. We report three appearances.

Next step: Euclid lensing prediction +5% in $r < 0.3R_{void}$, open for test.

Appendices (in Zenodo extended version)

A. Maximization of S/A for 5 Platonic solids
 B. Python code for void stacking
 C. L-BFGS-B optimization for δ_G

References

Franco Leon, M.A. (2026). Tetrahedral Emergent Gravity v7, Zenodo.
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 Franco Leon, M.A. (2025). IFCT Quaternion Solver v3.1, GitHub.
 DESI Collaboration DR1 BGS voids.

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